

AI-Assisted Machine Learning Project Report

Diabetes Prediction Using Machine Learning

1. Introduction

Diabetes is one of the most common chronic diseases in the world. It affects millions of people and can lead to serious health problems if it is not diagnosed early. Machine learning can help predict whether a person has diabetes by analyzing medical information. Early prediction allows doctors to provide timely treatment and improve patient care.

In this project, a machine learning system was developed to predict diabetes using patient medical records. The project followed a complete machine learning pipeline including dataset understanding, exploratory data analysis, data preprocessing, feature engineering, model training, model evaluation, and interpretation.

2. Dataset Understanding

The dataset used in this project is the **Pima Indians Diabetes Dataset**. It contains **768 patient records** and **9 columns**.

The target variable is **Outcome**.

0 = No Diabetes

1 = Diabetes

Therefore, this is a **Binary Classification** problem.

The numerical features are:

Pregnancies

Glucose

BloodPressure

SkinThickness

Insulin

BMI

DiabetesPedigreeFunction

Age

There are no categorical features in this dataset.

Based on medical knowledge, the most important features are **Glucose, BMI, Age, Insulin, and DiabetesPedigreeFunction** because they have a strong relationship with diabetes.

Two challenges in this dataset are:

- Some medical measurements contain **zero values**, which are not medically valid and should be treated as missing values.

- The dataset is slightly imbalanced because the number of diabetic and non-diabetic patients is not equal.

Artificial Intelligence assisted in generating preprocessing ideas, selecting machine learning models, interpreting feature importance, and explaining evaluation metrics.

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the dataset before building machine learning models.

The dataset dimensions, data types, missing values, duplicate records, statistical summary, class distribution, and correlation matrix were examined.

No duplicate rows were found in the dataset.

Although there were no missing values represented by **NaN**, several medical features contained **zero values**, which are unrealistic and were treated as missing values during preprocessing.

Several visualizations were created, including:

- Class Distribution
- Age Distribution
- Glucose Distribution
- BMI Distribution
- Correlation Heatmap
- BMI vs Outcome Boxplot
- Age vs Outcome
- Pregnancies Distribution

From these visualizations, it was observed that patients with higher glucose levels and higher BMI were more likely to have diabetes. Age also showed a positive relationship with diabetes risk. The correlation heatmap indicated that **Glucose** had the strongest positive correlation with the target variable.

4. Data Cleaning and Preprocessing

Several preprocessing techniques were applied to improve data quality before model training.

First, invalid zero values in **Glucose**, **BloodPressure**, **SkinThickness**, **Insulin**, and **BMI** were treated as missing values because these measurements cannot realistically be zero.

Median imputation was used to replace missing values because the median is less affected by outliers than the mean.

Duplicate records were checked and removed if present.

Outliers were handled using the **Interquartile Range (IQR)** method with capping to reduce the influence of extreme values while preserving all observations.

Since there were no categorical features in the dataset, no encoding was required.

Finally, **StandardScaler** was applied to standardize numerical features because they had different scales. Feature scaling improves the performance of algorithms such as Logistic Regression and Support Vector Machine.

These preprocessing steps produced a clean dataset suitable for machine learning.

5. Feature Engineering

Feature engineering was performed to improve the predictive ability of the models.

Several new features were created:

- BMI Category

- Age Group

- Glucose Level

- High Blood Pressure Indicator

- High Glucose Indicator

BMI was grouped into **Underweight**, **Normal**, **Overweight**, and **Obese** categories.

Age was divided into meaningful age groups.

Glucose values were categorized into **Normal**, **Prediabetes**, and **Diabetes** levels.

Binary features indicating high blood pressure and high glucose were also created.

These categorical features were converted into numerical values using **One-Hot Encoding**.

The performance of Logistic Regression was compared before and after feature engineering. The engineered features slightly improved the predictive performance and provided more meaningful information to the machine learning models.

6. Machine Learning Models

Three machine learning models were trained and compared.

The models were:

- Logistic Regression
- Decision Tree
- Random Forest

The dataset was divided into training and testing sets using an **80:20** split.

All models were trained using the same training data and evaluated on the same testing data to ensure a fair comparison.

Random Forest generally produced the highest performance because it combines multiple decision trees and reduces overfitting.

7. Model Evaluation

The models were evaluated using several performance metrics.

These metrics included:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix

Each model's confusion matrix was analyzed to understand the number of correct and incorrect predictions.

Among all evaluation metrics, **F1-score** and **ROC-AUC** were considered the most reliable because they provide a balanced evaluation of classification performance.

The model with the highest ROC-AUC and F1-score was selected as the best-performing model.

8. Best Model Selection

After comparing the performance of all three models, **Random Forest** was selected as the best model.

It achieved the highest overall performance in terms of Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Random Forest was able to capture complex relationships between features while reducing overfitting through ensemble learning.

Therefore, it was selected as the final prediction model.

9. Model Interpretation

Understanding model predictions is important, especially in healthcare applications.

Feature importance analysis showed that **Glucose** was the most influential feature for predicting diabetes.

Other important features included:

- BMI
- Age
- DiabetesPedigreeFunction
- Insulin
- Pregnancies

These features are medically meaningful because they are closely associated with diabetes risk.

Model interpretation increases the transparency of AI systems and helps healthcare professionals trust machine learning predictions.

10. AI-Assisted Development Reflection

Artificial Intelligence was used throughout this project to assist in code generation, data preprocessing, feature engineering, model development, and report writing.

Several prompts were used to generate Python code for Exploratory Data Analysis, preprocessing, feature engineering, model training, and evaluation.

The generated code was carefully reviewed, tested, and modified before being used.

Some AI-generated suggestions required correction because they assumed categorical variables existed in the dataset. These issues were fixed by manually reviewing the dataset and adjusting the code.

Overall, AI significantly reduced development time while improving productivity. However, human understanding and verification were necessary to ensure correctness.

11. Critical Discussion and Recommendations

Although the developed model achieved good performance, several limitations remain.

The dataset contains only **768 observations**, which limits the model's ability to generalize to larger populations.

The data represents a specific group of patients, so the model may not perform equally well for different populations.

Potential bias may exist because of class imbalance and demographic limitations.

Care was taken to avoid **data leakage** by separating the training and testing datasets before model evaluation.

Patient privacy is extremely important. Personal information should always be protected when developing medical AI systems.

This model should be used only as a **clinical decision support tool** and should never replace qualified medical professionals.

Future improvements include collecting larger datasets, using advanced ensemble models such as XGBoost, and improving model interpretability using SHAP values.

12. Conclusion

In this project, a complete machine learning pipeline was developed to predict diabetes using the Pima Indians Diabetes Dataset.

The project included dataset understanding, exploratory data analysis, data cleaning, feature engineering, machine learning model development, evaluation, and interpretation.

Among the trained models, **Random Forest** provided the best overall performance and demonstrated strong predictive capability.

The results show that machine learning can assist in the early detection of diabetes and support healthcare professionals in making informed decisions. However, AI should be used responsibly with proper validation, ethical considerations, and medical supervision before deployment in real-world healthcare systems.